



Reverse Polish and Infix Notation Calculator

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General Overview

What the application can do:

- Calculates expressions in Reverse Polish (Postfix) Notation.
 - E.g **3 5 +** and **4 3 +** as opposed to **3 + 5** and **4 + 3**. Strange notation, I know!
- Calculates expressions in good old Infix notation
 - E.g **10 + 2**, **5 + 3**, **8 + 6** **etc.**
 - This is your standard notation that everybody is familiar with.
- Takes as input expressions in either postfix or infix notation and prints result
- Converts expressions from infix to postfix notation. Reason why will become clear soon

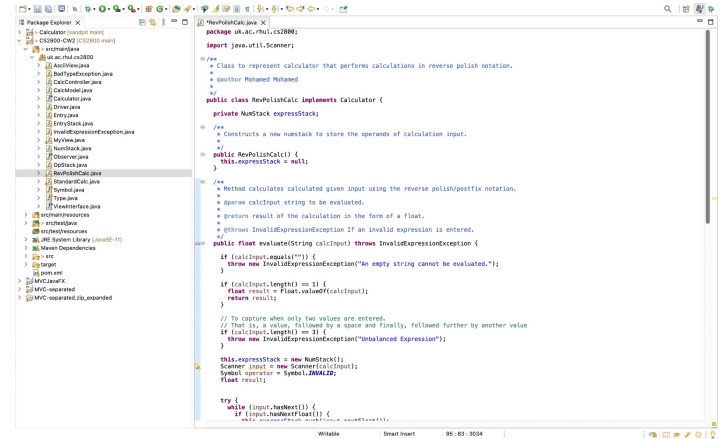
Reverse Polish Notation (Explained)

1. Operators are put after operands, **Postfix**. E.g $3\ 4\ +$ instead of $3 + 4$ as in the previous slide.
2. No need for brackets to enforce order of operations
 - E.g. $3 + 4 \times (2 - 1)$ would be written as $3\ 4\ 2\ 1\ -\ \times\ +$ in reverse polish notation
3. Easier for Machines to Interpret than infix expressions
4. Computers can evaluate Reverse polish notations much faster it much faster (Linear Time, $O(n)$)

Evaluating Expressions in Reverse Polish

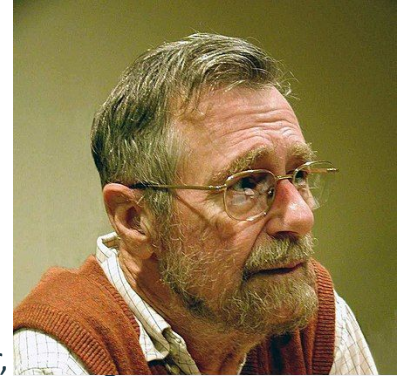
Algorithm works as follows:

- Scan user input from left to right and maintain a stack
- For each token in the string:
 - If token is an integer
 - push it onto the stack
 - If token is an operator,
 - pop the previous two items off the stack and do the calculation defined by the operator
- Finally, pop the last item off the stack and return the result



Evaluating Expressions in Infix Notation

- Evaluating Infix expression requires an awful lot of effort for computer to evaluate.
- How could this be made easier?
- Instead, given that we know computers can evaluate postfix expressions much faster, infix expression, we can convert it to reverse polish first, then evaluate that.
- This conversion is done using the **Shunting Yard Algorithm**
- This process of conversion from infix to postfix, in addition to the then evaluation of resulting reverse polish expression, still has a complexity of $O(n) + O(n) = 2 * O(n) = \mathbf{O(n)}$





Software Engineering Tools and Techniques Used

Tools and Techniques Used

- **TDD** (Test-driven Development)
- Design Patterns
- Checkstyle - Google Java Style
- Git/GitLab